

Linux Security Basics

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- S1.1. The following is an entry inside the `/etc/passwd` file. What is the user ID (numeric) of the user bob?

bob 的 user ID 是 2000

```
bob:x:2000:3000:SEED,,,:/home/bob:/bin/bash
```

- S1.2. What is the root user's user ID? Why does root have special privilege than normal user? Is it because its user name is root?

- S1.3. Alice belongs to the `abc` group. What permission does Alice have on the following file?

```
-rwxr--r-- seed abc 1802 Feb 6 11:39 xyz
```

Alice 只有读xyz文件的权限

- S1.4. What will be the file `xyz`'s permission after running the following command?

```
$ chmod 543 xyz r-xr--rw-
```

- S1.5. If the `umask` value is `0427`, when we create a non-executable file, what will be its permission?

- S1.6. The account `bob` is a normal user account. The root user wants to grant `bob` the power to run commands using the superuser privilege, but without giving `bob` the password of the root account. How can the root user achieve this?

把 bob 加入到 sudoer 组

- S1.7. We are allowed to run commands using the superuser privilege (via `sudo`). How do we get a root shell?

使用 `sudo su` 或者 `sudo bash` 命令

- S1.8. By exploiting a vulnerability in the system, an attacker gets a chance to modify any arbitrary file in the system. What file can the attacker modify, such that after modifying the file, the attacker can gain the root privilege on the system. Please name two files. Assume that the attacker has a normal-user account on the system.

`passwd` 和 `shadow`

- S1.9. What are the main objectives of the POSIX capabilities? 给程序设置细粒度特权

- S1.10. We run the following sequence of commands. What are the results at ① and ②?

```
$ cp /usr/bin/cat mycat
$ mycat /etc/shadow ① 1 permission denied
$ sudo setcap CAP_DAC_READ_SEARCH=ep mycat
$ mycat /etc/shadow ② 2 permission allowed
```

- S1.11. The `/etc/passwd` file is called password file, but in reality, it does not contain account passwords. Please explain why?

- S1.12. What is the purpose of the salt in the shadow file? to prevent two equal password get the same hash value

root 用户的id 是 0, 因为这个 0 的id表示超级用户, 用户名 root 与特权无关, 甚至可以改为其他名字, 但只要id为 0, 就具有特权, 是超级用户。

`umask: 100,010,111`

`~umask: 011,101,000,`

`perm: -w-r-----`

because it can be read by anyone, so the password is not stored here but instead stored in shadow file which is only readable/writable to root for security